



Predicting Market Volatility: An Ensemble Approach for Enhanced India VIX Prediction

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Abstract

Asset allocation relies heavily on accurate volatility predictions due to the correlation between volatility and financial risk. The India VIX index is a gauge of the market's risk assessment, and a precise estimation of India VIX prices can benefit financial risk mitigation. This research presents an ensemble deep learning framework that integrates long short-term memory (LSTM) and gated recurrent unit (GRU) structures to predict India VIX values. Hyperparameter tuning was carried out using random search to optimize performance, guaranteeing an effective exploration of the parameter range. Ensuring generalisability across various market situations, the proposed ensemble model was thoroughly examined using various measures, and robustness was proven using five-fold cross-validation. Experimental findings indicate that the ensemble LSTM-GRU approach surpasses independent deep learning methods, successfully predicting volatility patterns even amid market instability. The suggested model's adaptability was further assessed using stress testing in unstable market circumstances, validating its reliability and consistent prediction performance. The Diebold–Mariano test confirmed that the ensemble model greatly surpassed all benchmark models, hence validating the results' reliability. This research promotes the application of ensemble learning in finance, offering a means to improve forecast accuracy in evolving market scenarios.

Keywords India VIX · LSTM · GRU · Deep learning · Ensemble learning · Volatility prediction

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1 Introduction

Volatility in financial markets elicits diverse reactions from traders and investors. Some view it as potential for profit, while others regard it as an imminent danger and choose to liquidate their assets. Aside from influencing investment decisions, financial market volatility provides insight into the economy's health. In the contemporary globalized landscape, heightened volatility signifies worldwide risk. The volatile nature of particular stocks may stem from the projected potential for expansion of the firm or its industry. Corporate policy or headlines might potentially set it off as well [2]. Established in 2008 by India's National Stock Exchange (NSE), the India VIX measures the implied volatility of options on the NIFTY 50 Index. The index calculation approach is derived from the Chicago Board Options Exchange (CBOE) Implied Volatility Index (VIX) with authorization from the CBOE. The India VIX is determined by utilizing the IV of options traded on the NIFTY 50 Index [12].

Predicting volatility has consistently posed the greatest issue for financial modelers. This is due to the fact that volatility data is characterized by a great deal of complexity, nonlinearity, and spatial fluctuation. There are primarily two methods for predicting volatility in the field: the data-driven method, sometimes known as the black-box method, and the stochastic method. Without assuming anything and depending solely on training techniques to reduce training error, the data-driven strategy seeks to derive the best fit from input data. Nevertheless, the stochastic methodology relies on the probability density function (pdf) [14].

A stochastic framework for predicting probability distributions of desired results by including variation at random in multiple variables over the course of time. To illustrate, real financial data sets are frequently modeled using the stochastic GARCH model, which offers a straightforward depiction of the primary statistical properties of return sequences for various assets. Professionals frequently link the data-driven architecture technique to mathematical modeling and computational intelligence. Despite several data-driven methodologies, artificial neural networks (ANNs) and support vector regressors (SVRs) are frequently utilized. Each neuron in an artificial neural network (ANN) receives information and produces a response based on its activation function. With the help of a supervised algorithm for learning, ANN can figure out how to map variables to outcomes by sampling from a training set. Subsequently, the trained artificial neural network is employed to predict outcomes for the test dataset [10].

In recent decades, deep neural networks (DNN)—an advanced branch of ANNs—have attained significant appeal owing to their efficacy in predictive tasks. “DNNs” refer to an approach to acquiring data representations with several degrees of complexity by means of numerous computational levels. Computer vision, computational finance, robotics, speech recognition, and natural language processing are just a few areas where DNN has been demonstrating encouraging results. As recurrent neural networks (RNNs) have evolved across multiple academic domains, scholars have begun utilizing RNNs to address regression challenges involving sequential data. A significant issue faced during the training of networks utilizing RNN is their inability to maintain long-term input dependency [21]. Derived from recurrent neural networks (RNNs), LSTM and GRU address the issue of gradient vanishing during RNN training

and serve as an efficient instrument for long-term forecasting. Recently, long-short-term memory and gated recurrent units have acquired prominence in the deep learning world for predicting time series. A number of scholars endeavor to integrate CNN-LSTM, CNN-GRU, CNN-BiLSTM, CNN-BiGRU, etc. to create hybrid models for medical and stock price forecasting [17, 19], among other applications. A substantial body of empirical evidence indicates that the singular neural network prediction model frequently struggles to effectively forecast financial market prices due to the influence of trade discontinuities.

Ensemble learning denotes a methodological approach that amalgamates multiple baseline models to construct a singular, more potent model than its individual components. Moreover, ensemble learning mitigates the danger of overfitting due to the heterogeneity of foundational models. Ensemble learning has been effectively utilized across diverse sectors and domains, surpassing the performance of individual models. A number of ensemble methods exist, each with its own unique approach to training and combining baseline models. The predominant ensemble techniques comprise averaging, bagging, random forest, stacking, and boosting [2]. This study proposes a framework that ensembles two widely known deep learning frameworks—LSTM and GRU.

Although LSTM and GRU-based architectures are well known in the area of time series forecasting, their synergy within a stacking-based ensemble structure has not been explored enough, especially in an environment of developing markets such as the Indian market. In fact, using LSTM and GRU together in an ensemble approach offers a new approach to the volatility forecast problems. An ensemble LSTM-GRU framework is expected to improve the accuracy of the predictions, increase the model's resilience, and give a broader perspective on the market. It is widely assumed that ensemble techniques are less prone to overfitting and produce higher accuracy by canceling the noise of component models. The contribution of the research to the literature is the ability to incorporate LSTM and GRU with the assistance of the meta-learning method, as well as compare the resulting ensemble with the traditional econometric computations (ARIMA, GARCH) and the benchmarking to individual deep learning baselines. Moreover, an emphasis on Indian VIX addresses the literacy deficit that exists in the majority of the previous research focusing on developed markets. Comparable with the literature review, when applied to India VIX prediction, the LSTM-GRU ensemble will yield higher accuracy than the individual models because the former is more adept at recognizing the interconnections within the financial data and delivers more accurate forecasts during periods of volatility. Extensive tests have been conducted to demonstrate the efficacy of the suggested ensemble model under the experiment section. To validate the performance of the model, a proper five-fold cross-validation scheme is adopted to ensure a sturdy evaluation under various market scenarios. All these aspects predetermine the methodological and applied originality of the given work. This research presents the subsequent findings:

- An LSTM-GRU ensemble model is proposed in the study to predict the closing price of the India VIX Index.
- Utilized random search for choosing the best hyperparameters in order to enhance the output of the ensemble model.

- Employed numerous metrics of evaluation to critically evaluate the model, suggesting the overall potential of the model in its predictive aspect.
- Used a five-fold cross-validation approach in order to minimize the potential for overfitting and increase the model's reliability when tested on new data samples.
- Ensured the suggested model's robustness through stress testing in high-volatility market conditions, with minimum performance deterioration.
- Conducted the Diebold–Mariano statistical test, which validated that the performance enhancements of the ensemble model relative to baseline models are statistically significant at the 5

The remaining sections are organized as follows: Section 2 gives a review of existing studies; Section 3 describes the models and methods used in the study; Section 4 first discusses the dataset and experimental design. Further, results and Robustness of models are also discussed in Section 4. Lastly, Section 5 concludes the study with some future ideas.

2 Literature Review

The forecasting of financial variability, particularly via indices such as India VIX, has been a persistent focus in finance research because of its vital importance in evaluating market risk and informing investment strategies. Using covariance, researchers began studying volatility in the early 1950s with the aim of predicting changes in stocks. A multitude of studies have employed the GARCH model for evaluating volatility, as the GARCH series of models is recognized for producing superior outcomes. With the popularity of deep learning, such as recurrent neural networks (RNNs), long short-term memory (LSTM), and gated recurrent units (GRU), these algorithms have become the mainstream to forecast volatility because of their ability to model long dependencies in time series data.

Xiong et al. [20] utilized 27 parameters to predict the volatility of the *S&P 500* for their brief investigation. The 25 domestic trends, in conjunction with previous volatility and returns, functioned as variables for LSTM RNNs. However, only one-layer LSTM RNNs were employed, and their predictive performance was only slightly better than that of the GARCH model when evaluated with root mean square error (RMSE) as the measure.

Liu [10] utilized LSTM to anticipate the volatility of the *S&P500* and *AAPL*, claiming that LSTM can integrate vast amounts of primary data along with several hidden layers and neurons on a GPU to achieve precise predictions for extended sequence data.

Ballestra et al. [1] utilized an artificial neural network to create a framework distinguished by minimal prediction error and significant profitability for predicting future VIX movements. This is an intra-day approach that emphasizes open-to-close returns, utilizing non-lagged external influences derived from the data and information available to traders for investment decisions.

Dixit et al. [5] employed an ANN model to predict daily change in directional fluctuations in the India VIX, achieving a precision of up to 61% based on technical indications extracted from price data.

Chaudhuri and Ghosh [2] employed artificial neural network models utilizing various backpropagation techniques to forecast volatility in India's equity markets, specifically targeting the swings of NIFTY and gold gains. Utilizing training data from 2013 to 2014, their findings indicate that the model effectively predicted volatility for 2015. Nonetheless, the predictive accuracy of market volatility in 2008 was diminished while utilizing the identical training data sets.

Kumar et al. [7] examined the impact of profits on the NIFTY Index, foreign portfolio investment (FPI) in India, the *S&P* 500 Index, and the CBOE VIX Index on the India VIX utilizing the ARIMA-GARCH model. Research showed that the India VIX Index was affected by the NIFTY Index, the FPI in India, the *S&P* 500 Index, and the CBOE VIX Index. They specifically noted that the India VIX Index has solely diminished due to heightened funding for FPI.

Jung and Choi [6] employed an autoencoder-LSTM to anticipate future FXVIX values utilizing training information from 2010 to 2019, asserting that their findings can positively assist investors in decision-making.

Chen et al. [4] introduced an innovative hybrid deep learning approach for volatility prediction and demonstrated its superiority over current models through the Friedman test.

Choi and Shin [3] suggested an ensemble circuit including three CNN-BISTM networks to improve the precision of implied volatility forecasts.

Prasad and Bakhshi [12] and Prasad et al. [13] forecasted the daily fluctuations of India VIX utilizing multiple machine learning methods.

Recent research has also highlighted the efficacy of hybrid models that integrate statistics and deep learning methodologies for volatility forecasting. LSTM-ARIMA frameworks have garnered interest for their capacity to identify both linear patterns (by ARIMA) and nonlinear dependencies (via LSTM), resulting in enhanced forecasting precision. These models utilize the advantages of conventional time series analysis and deep learning to tackle the intricate characteristics of financial data. Numerous empirical studies have indicated that these hybrids surpass independent LSTM or ARIMA models, especially in volatile market conditions, underscoring their practical significance in financial forecasting endeavors [8].

Roszyk and Ślepaczuk [15] applied four models to forecast *S&P* 500 volatility: the GARCH model, LSTM, a hybrid LSTM-GARCH, and an extended hybrid model incorporating the VIX index. Using daily data from 2000 to 2023, the findings showed that hybrid LSTM-GARCH models, particularly those including VIX as a sentiment indicator, consistently outperformed the standalone GARCH model.

Liang and Ismail [9] proposed a hybrid VIX forecasting model by decomposing the time series using complete empirical mode decomposition (CEEMD) into intrinsic mode functions (IMFs). Stationary IMFs were modeled with ARIMA, while non-stationary components used machine learning regressors. The final prediction was the sum of all IMF forecasts. Hyperparameters were optimized using Optuna to minimize MSE. Their results showed that the hybrid model, especially AdaBoost, outperformed baseline models during high market volatility.

Though many methods for forecasting volatility have been proposed in the literature, limited research has been conducted to use ensemble methods for the India VIX. Previous research work involves the use of LSTM or GRU in isolation or conventional models, with little emphasis on hybrid models that can provide better predictive capability by combining results from individual models. To fill this research gap, this research presents an ensemble LSTM-GRU model for the prediction of India VIX. The proposed model successfully integrates the strongest aspects of the two architectures, and it can capture various levels of temporal dependencies and, thus, can have better prediction accuracy compared to traditional models. Thus, the addition of this study lies in providing an efficient and correct method of predicting the volatility of the Indian financial market concentration, namely India VIX.

3 Models and Methods

The behavior of volatility in financial markets is contingent on the effect of investor behavioral components, macroeconomic news, and systemic risk factors, and is frequently nonlinear and time-dependent. Their intricate nature demands forecasting models that transcend the common assumptions of an econometric model. Specifically, financial time series like India VIX have long-run dependence, asymmetric shock response, and sudden regime switching, which become challenging to model using linear or memoryless models.

Recurrent neural networks (RNNs), in particular, have proven a potent solution in that regard. RNNs, unlike traditional models, work on sequences; hence, they are appropriate for time series predictions. Among these, long short-term memory (LSTM) and gated recurrent unit (GRU) patterns are the most well known, recognized by their capability to remember historical patterns and their flexibility to transform data structures with time. These models provide a natural candidate to model the implied volatility, which is the manifestation of the changing risk view in the market caused by recent events, as well as behaviors far in the past. Their capability to capture lag effectiveness, volatility clustering, and nonlinearity dependence linkage is a cost-effective basis on which forecasting operations can be based. In the next two subsections, the LSTM and GRU models, which will be used in this study, are outlined. These models are subsequently combined into a stacked ensemble model that helps maximize their capabilities toward enhanced prediction performance and resilience.

3.1 Long Short-Term Memory

LSTM, a sophisticated gated memory unit, is designed to mitigate the issue of vanishing gradients that impede the efficacy of traditional RNNs, thereby providing a more resilient alternative for sequential processing of data in machine learning exploration. A cell state, an output layer, a hidden layer, and an input layer make up the system. The pivotal component of the LSTM design is the cell state, which progresses through a sequence with exclusively linear interactions, preserving the integrity of the information flow. LSTM can modify or eliminate cell state information through

its gating mechanism. This selective data transmission technique includes a sigmoid layer, a hyperbolic tangent layer, and a pointwise multiplication operation. Figure 1 illustrates the LSTM architecture at a particular time t , highlighting its configuration for modeling sequential input. Four gates are emphasized: output (o_t), change (c_t), input (i_t), and forget (f_t), each demonstrating their specific functions at time t [16].

3.2 Gated Recurrent Unit

An improved version of the LSTM model, the gated recurrent unit (GRU), uses fewer processing cycles. To streamline the process, GRU integrates the short-term and long-term data from LSTM into a singular vector. In contrast to LSTM's four-gate configuration, the GRU architecture streamlines the gating mechanism into three elements: the reset gate (r_t), the update gate (u_t), and the change gate ($\tilde{h}_t$). Remarkably, the update gate in GRU performs a dual purpose, similar to the input gate and forget gate in LSTM. This design variant provides a sophisticated method for managing data flow inside the network, which may influence performance on different tasks in machine learning studies [18] (Fig. 2).

3.3 Ensemble Learning

In this current study, an ensemble of two models was created to increase the probability of accurate prediction of India VIX prices. In the context of the proposed ensemble framework, LSTM and gated recurrent units (GRU) are used as the basic learning models. This ensemble combines the advantages of both LSTM and GRU in terms of temporal dependency and diversification needed in sophisticated time series forecasting.

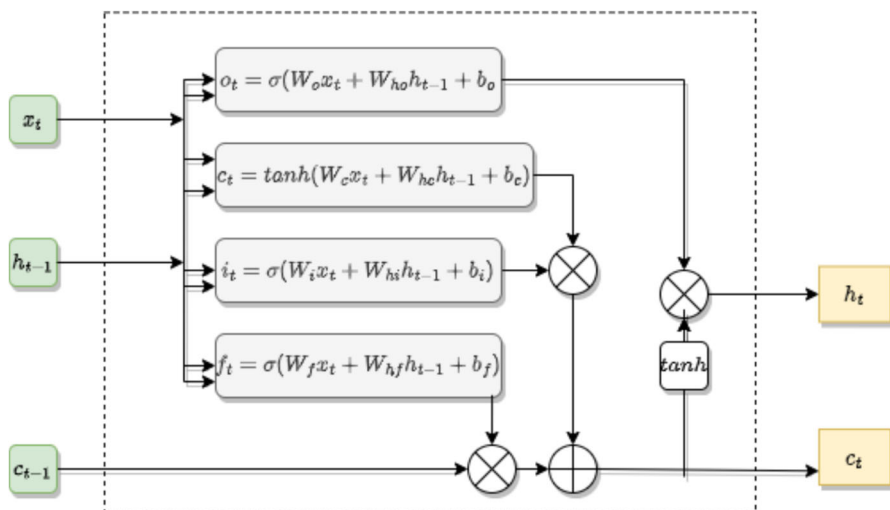


Fig. 1 An LSTM unit

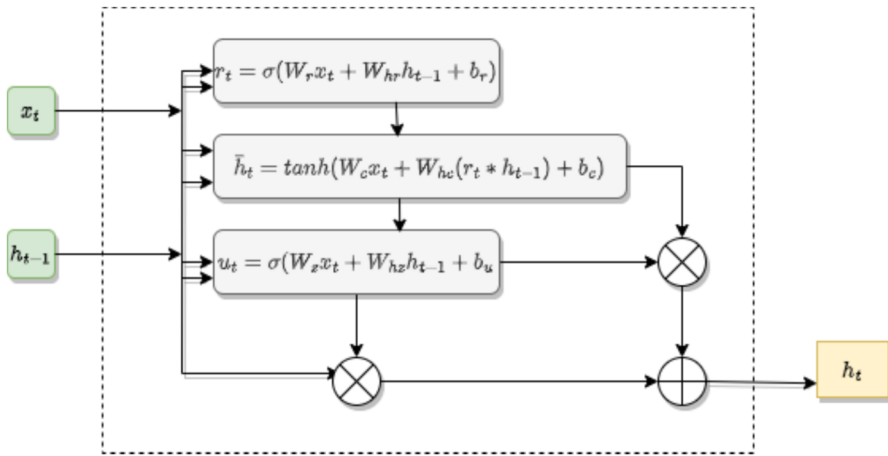


Fig. 2 A GRU unit

Ensemble methodology is to build individual base learners where each learner performs the task of learning and prediction independently and builds a meta-learner where multiple predictions are integrated. Here, linear regression is used as the meta-learner along with the predictions of LSTM and GRU. This stacking approach is described in Fig. 3, which presents the workflow of our ensemble learning model. Since the meta-learner has the ability to sum up the weights of each base learner's predictions, the meta-learner can take advantage of the features learned by the individual base learners, thereby providing increased robustness and higher accuracy to the model. Unlike conventional methods of ensemble approaches like bagging and boosting that are usually implemented on tabular or independent data, the proposed stacking-based ensemble is particularly developed to exploit the sequential dependence and dynamic time series that are synonymous with the financial time series data [11].

The ensemble architecture in this research is based upon a two-tier stacking architecture. The initial level consists of two simply trained base models, LSTM and GRU,

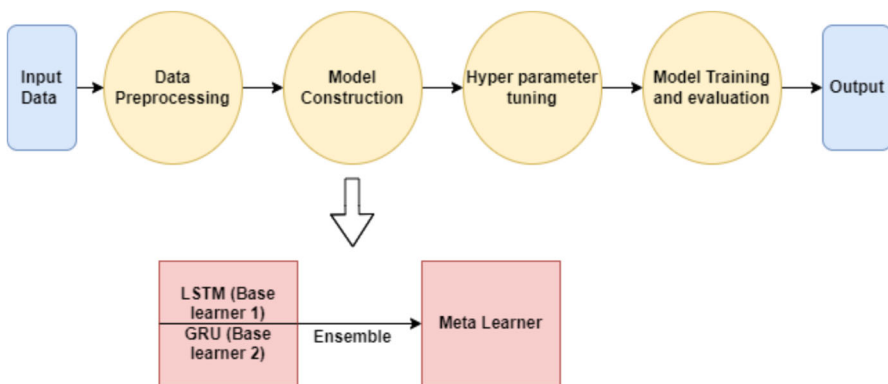


Fig. 3 Framework of methodology

that will be trained on the same input features but separately. All these models give their individual forecasts of the target variable. As a second level, the predictions of the two base models are used to train a meta-learner, a linear regression model, which is to be used later. The meta-learner is trained to put the best weights on the output of each of the base models to make the final forecast. The method has the benefit of merging the capabilities of the two recurrent architectures, so that the resulting prediction outputs the most informative features of both.

3.4 Evaluation Metrics

- **Root mean squared error (RMSE):** It assesses the degree of deviation of the prediction error from the mean, highlighting outliers that exceed the typical range. Reduced real RMSE values indicate enhanced predictive accuracy.
- **Determination coefficient:** It denotes the percentage of volatility in the dependent variable that may be forecasted from the independent variables. A higher score indicates enhanced performance for the model.
- **Mean absolute error:** It calculates the average absolute divergence between expected and real outcomes, functioning as a measure of predictability.
- **Mean squared error:** calculates the mean of the squared deviations between the anticipated and actual outcomes.
- **Value at risk (VaR):** It is a statistical metric commonly employed in finance to assess the possible loss of an asset or portfolio over a defined time frame, contingent upon a specific confidence level. This study calculates $\text{VaR}_{95\%}$ to ascertain the greatest anticipated loss in VIX price forecasts under standard market conditions.
- **Expected shortfall (ES):** It is also known as conditional value at risk (CVaR) and enhances value at risk (VaR) by detailing the extent of losses that surpass the VaR limit. Specifically, $\text{ES}_{95\%}$ denotes the mean of the most adverse 5% of forecast errors, so encapsulating tail risk that exceeds the probabilistic loss threshold signified by VaR.

4 Experiment and Results

4.1 Dataset Description

This research relies on publicly accessible data. The daily statistics of the India VIX were obtained officially from the NSE of India website. The research time-frame spanned from January 2014 to October 2024, encompassing sections for open, high, low, close, change, and percent change. Following that, the data underwent pre-processing, and the feature and target variables were created. Prior to inputting data into the framework, it is essential to modify it in order to ensure it is unitless or conforms to a consistent unit, since deep learning frameworks have sensitivity to input data, making data scaling a critical step in preliminary processing. For this purpose, the MinMax Normalization technique is used to scale the data into a 0 to 1 range. The

dataset was split into training, testing, and validation subsets with a ratio of 60:25:15, respectively.

4.2 Experimental Design

All the experiments were performed on Google Colab using the latest Python version and GPU computing. The suggested strategy incorporates LSTM and GRU units inside an ensemble architecture, enabling the framework to observe both short- and long-term relationships in the time series dataset. In order to mitigate the potential repercussions of overfitting during neural network training, early stopping criteria have been adopted. It enables us to designate a substantial number of epochs and terminate training once the model's effectiveness ceases to improve on the validation data. Subsequent to model construction, we proceed with a hyperparameter tuning procedure wherein each model determines its optimal set of hyperparameters using various methods. The hyperparameters were selected using the random search since it unveils a large hyperspace of values without necessitating vast computation. The hyperparameters that were allowed to be tuned include the number of LSTM and GRU units, rate of dropout, learning rate, and batch size. Table 1 shows the parameters selected for the model.

4.3 Results

Upon determining the ideal hyperparameters and completing model training, we subsequently evaluate the performance metrics of the proposed model against other models. The proposed LSTM-GRU model showed high predictive accuracy in the overall

Table 1 List of selected parameters

Model	Parameter	Range	Value
LSTM	Units	16, 32, 50, 64, 100, 150, 200, 256	64
	Layers	1–5	2
	Dropout	0.1–0.9	0.1
	Dense		1
	Optimizer	Adam, Nadam	Nadam
	Batch size	16, 32, 64, 128	32
	Learning rate	0.1–0.00001	0.001
GRU	Units	16, 32, 50, 64, 100, 150, 200, 256	100
	Layers	1–5	2
	Dropout	0.1–0.9	0.1
	Activation	Relu, tanh	Relu
	Optimizer	Adam, Nadam	Adam
	Batch size	16, 32, 64, 128	64
	Learning rate	0.1–0.00001	0.001

test data set, and it depicted the trends of the India VIX price movements. In correspondence with this, Fig. 4 presents the comparison of predicted values to the actual, indicating that the former successfully reflects the volatility of the values detected in the ground-truth.

Some of the evaluation metrics that have been tested are mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and determination coefficient (R^2). Table 2 delineates the performance measures of all assessed models, contrasting traditional and hybrid methodologies. The suggested LSTM-GRU ensemble attains the minimal RMSE (0.01300) and MAE (0.00822), as well as the maximal R^2 score (0.94338), signifying exceptional prediction accuracy. It also exhibits the lowest risk exposure, as indicated by the most advantageous expected shortfall (ES) and value-at-risk (VaR) metrics. Conversely, the ARIMA-GARCH model demonstrates substantial underperformance, characterized by elevated error rates and a negative R^2 , indicating inadequate fit. The hybrid ensemble comprising LSTM and ARIMA-GARCH demonstrates enhanced performance compared to ARIMA-GARCH alone, although it remains inferior to the suggested deep learning-based ensemble. These findings validate the efficacy and practical significance of the LSTM-GRU methodology for precise volatility forecasting in financial markets.

The ensemble LSTM-GRU model's ability to predict India VIX values accurately is attributable to its architecture, which leverages both LSTM's strength in handling long-term dependencies and GRU's simplified structure, enabling faster convergence and improved generalization. The model was tested during periods of increased market volatility, as reflected in the VIX spikes, where it successfully captured the sharp fluctuations, demonstrating its robustness during turbulent market conditions. The strong predictive accuracy of the model can aid investors, financial institutions, and policymakers by providing more reliable volatility forecasts, thus improving decision-making in risk management and investment strategies.

The suggested configuration of ensemble LSTM-GRU shows quite promising results in predicting daily volatility during a training period, but such an approach raises other concerns when applied to real-life financial settings with time constraints. Specifically, high-performance trading (HFT) applications require ultra-fast latency,

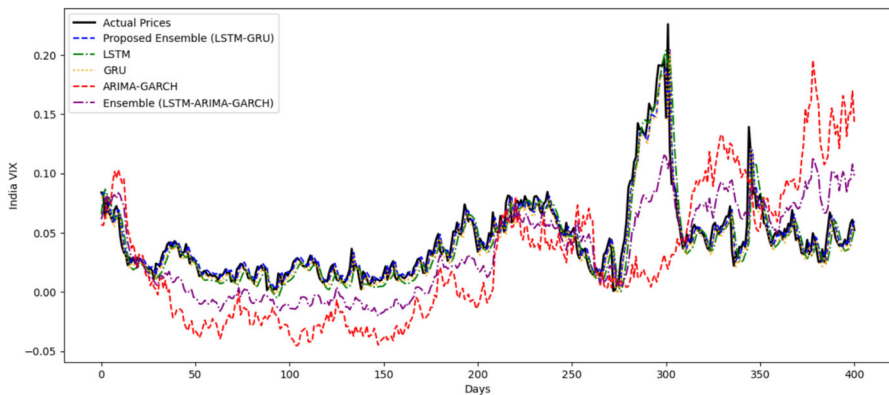


Fig. 4 Actual vs predicted graph on test dataset

Table 2 Performance score of all models

Model \ Metrics	RMSE	MAE	R^2	MSE	VaR (95%)	ES (95%)
LSTM	0.01498	0.01114	0.92476	0.00022	0.00000	-0.1616
GRU	0.01357	0.00888	0.93832	0.00018	0.00000	-0.1527
Proposed ensemble (LSTM-GRU)	0.01300	0.00822	0.94338	0.00017	0.00000	-0.1472
ARIMA-GARCH	4.23791	3.44349	-1.77732	17.95990	-6.70544	-8.14470
Ensemble (LSTM+ ARIMA-GARCH)	2.26211	1.83347	0.20868	5.11715	-3.28293	-4.13861

throughput, and computation effectiveness that cannot always be met by deep learning architectures, particularly in the recurrent schemes. This study is designed to work with daily data and is not aimed at the HFT deployment, but the training and inference time of the proposed model is already optimized to be practically feasible.

4.4 Robustness Check

4.4.1 Cross-validation

The model was validated with a five-fold cross-validation scheme to guarantee a strong assessment and prevent overfitting to one train-test combination. The whole dataset was divided into five chronological segments of the same size. Four segments acted as the in-sample (training) set in every fold, and the rest acted as the out-of-sample (testing) set. This was done five times, wherein each fold was utilized once as the testing period. This type of arrangement subjects the model to a variety of market conditions, allowing it to measure its performance under different volatility regimes and return distributions. The metrics used to evaluate the models presented are averaged across the five folds; therefore, they are more generalizable and reliable in reflecting the model's capabilities to make predictions. In Table 3 below, the overall

Table 3 Average performance score of all models (cross-validation)

Model \ Metrics	MSE	RMSE	R^2	MAE	VaR (95%)	ES (95%)
Proposed ensemble (LSTM-GRU)	0.000632	0.025138	0.916238	0.015342	-0.2912	-0.3846
GRU	0.000801	0.028291	0.897614	0.017204	-0.3175	-0.4211
LSTM	0.001143	0.033790	0.845109	0.021584	-0.3397	-0.4582
Ensemble (LSTM+ARIMA-GARCH)	0.004925	0.070191	0.313078	0.066814	-0.5924	-0.7428
ARIMA-GARCH	8.613762	2.933240	-1.138475	2.651003	-5.3149	-6.2094

average accuracy of the five folds is presented, demonstrating quite robust performance of the model. In general, there was a small standard deviation between the folds in all the evaluation metrics, making the average of the MAE, MSE, RMSE, and R^2 stable. This consistency goes on to demonstrate the strength of the model in generalizing well to unseen data, as reconfirmed by low variance in the metrics across the folds.

4.4.2 Stress Testing

To assess the resilience of the proposed model under harsh market situations, we performed stress testing utilizing data from the volatility rise produced by COVID-19 between February and May 2020. Figure 5 depicts the model's predictions in comparison to the actual India VIX levels throughout this timeframe. The LSTM-GRU Ensemble model did a great job of following the observed surge in volatility, precisely matching both the peak and the drop in VIX values that followed. Conversely, conventional models like Hybrid ARIMA-GARCH and the Ensemble LSTM+ARIMA-GARCH exhibited substantial underperformance, remaining stagnant or misaligned with actual market dynamics.

The performance metrics shown in Table 4 substantiate these conclusions. The LSTM-GRU Ensemble attained the minimal RMSE (5.81) and MAE (4.19), alongside the maximal R^2 score (0.8861), signifying exceptional predictive precision. Furthermore, its Value-at-Risk (VaR) and Expected Shortfall (ES) at the 95% level were moderate, indicating a balanced tail risk during crisis scenarios. Standalone deep learning models (LSTM and GRU) exhibited commendable performance, albeit inferior to that of the ensemble. The Hybrid ARIMA-GARCH and Ensemble LSTM+ARIMA-GARCH models displayed negative R^2 values and significantly elevated error metrics, underscoring their inadequacies in modeling nonlinear volatility dynamics during periods of turbulence.

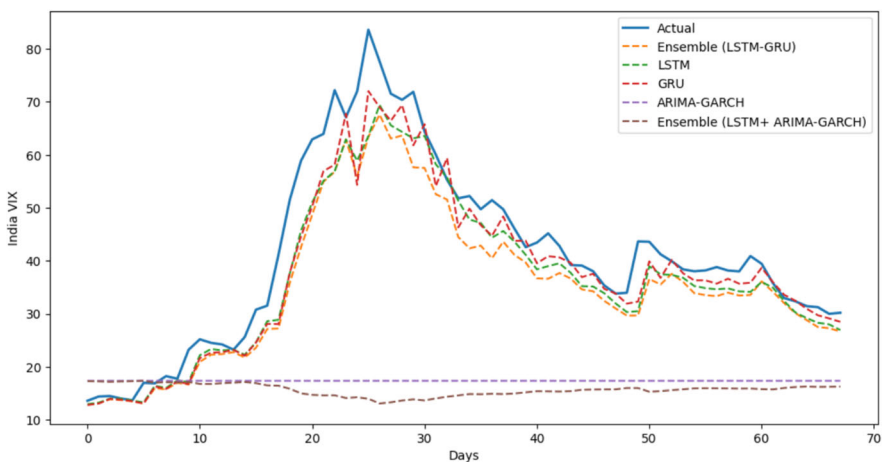


Fig. 5 Model predictions during the COVID-19 stress period (Feb–May 2020)

Table 4 Stress period performance metrics

Model \ Metrics	RMSE	MAE	MSE	R^2	VaR (95%)	ES (95%)
LSTM	6.0013	4.2410	36.0162	0.8784	−0.2935	−0.5097
GRU	6.5709	4.9458	43.1770	0.8542	0.2485	0.0565
Proposed Ensemble (LSTM-GRU)	5.8081	4.1862	33.7345	0.8861	−0.0103	−1.8329
ARIMA-GARCH	29.2584	24.1756	856.0537	−1.8905	−2.9262	−3.4364
Ensemble (LSTM+ARIMA-GARCH)	30.6854	25.2941	941.5960	−2.1793	−2.7549	−3.3057

The results highlight the effectiveness of the suggested multivariate ensemble method in improving prediction resilience under financial stress events.

4.4.3 Statistical Significance

To assess the statistical significance of the predictive performance enhancements provided by the proposed ensemble model, we utilized the Diebold–Mariano (DM) test. The DM test is a statistical hypothesis test intended to evaluate the predictive accuracy of two rival models. It evaluates the null hypothesis, asserting that both models possess equivalent prediction accuracy, in contrast to the alternative hypothesis that one model outperforms the other. This study employed the mean squared error (MSE) as the loss function for the test. A positive DM statistic signifies that the ensemble model produces a lower MSE than the benchmark model, while a p -value within the 0.05 significance threshold results in the rejection of the null hypothesis, suggesting that the performance disparity is statistically significant.

Table 5 displays the DM test outcomes contrasting the ensemble model with LSTM, GRU, Hybrid ARIMA-GARCH, and an ensemble LSTM+ARIMA-GARCH model. In every instance, the DM statistics are positive, and the p -values are well below the 0.05 threshold, frequently nearing zero. The results demonstrate that the ensemble model regularly and considerably surpasses each rival model regarding forecasting accuracy. The findings not only confirm the strength of the proposed framework but

Table 5 Diebold–Mariano test results (loss function: MSE)

Model vs Ensemble	DM Statistic	p -value
LSTM	3.6577	0.000254
GRU	9.4926	< 0.0001
ARIMA-GARCH	12.4371	< 0.0001
Ensemble LSTM-ARIMA-GARCH	10.8482	< 0.0001

Notes: Positive DM statistics indicate that the ensemble model in the comparison has lower mean squared error than the competing model. p -values < 0.05 indicate statistical significance

also illustrate its superiority over both independent deep learning architectures and hybrid statistical-deep learning methods.

5 Conclusion and Future Work

In this research, an ensemble LSTM–GRU model was employed for predicting India VIX prices to solve the problem of complicated non-linear patterns inherent in the financial volatility data. Using the features of LSTM and GRU structures, the errors in the time series data were corrected, and both short- and long-term dependencies were accurately predicted, resulting in higher efficiency compared to traditional and isolated deep learning models. The performance of the ensemble model was evaluated using matrices such as Mean Absolute Error (MAE), mean squared error (MSE), root mean squared error (RMSE), and coefficient of determination (R^2), and cross-validated with internal five-fold cross-validation to increase reliability.

The results show how the proposed ensemble LSTM-GRU model has improved not only the accuracy of the forecast but also offered a valid approach for predicting India VIX during a volatile market. More precise VIX forecasts help to make risk budgeting, reallocation of assets, and derivative-based hedging schemes more responsive. Volatility targeting and statistical arbitrage strategies are used by institutional investors, who could utilize these forecasts to trade in an automated system in real-time exposure moderation. Policymakers and regulators could also utilize timely volatility information to monitor systemic risk, calibrate margin requirements, or implement circuit breakers in turbulent times. The high degree of precision and stability shown by the ensemble model strengthens its use in decision-making in informed, data-driven approaches in all financial sectors. In addition, hyperparameter tuning using the random search method guaranteed that the chosen model configuration provided very good predictive accuracy while minimizing time consumption.

Nevertheless, there are a number of shortcomings that need to be noted. The first methodological weakness was that the research concentrated solely on India VIX as a target variable. Though such a decision corresponds to the intent to track volatility in the Indian market, it limits the application of the model to the universal financial context in the global view. The addition of more indices, like the CBOE VIX or Euro VIX, would also aid in the assessment of the ability of the model to be transferred across geographies. Second, hyperparameter tuning was carried out through random search and produced stable results, but the sensitivity analysis, to a large extent, was not carried out. This hampers interpretability as to the effects of particular choices of parameters on the model performance. Third, the existing model does not consider global or macroeconomic risks as a factor in the forecasting process, which would be useful in supplementing the context of forecasting and also enhancing accuracy, particularly in a period of systemic shocks.

Future researchers can overcome the limitations above by expanding the data set with several volatility indices, considering macroeconomic indicators or high-frequency trading data, and comparing them to state-of-the-art models such as Transformer or Informer models. Further transparency and faithfulness of the stakeholders might be increased with further explanation processes, like SHAP values or

attention mechanisms, being added to the model. It would be useful to see the performance and implementation of the model in live trading or risk management systems in real time.

Author Contribution P.S.-Experimental work, Problem finding, Proof Reading; A.S.-Drafting, Writing; M.J.-Supervision, Proof Reading; C.K.V.- Supervision, Proof Reading.

Data Availability Data will be made available on request.

Declarations

Ethics Approval Not applicable. This study did not involve human participants or animals.

Consent to Participate Not applicable

Consent for Publication Not applicable

Conflict of Interest The authors declare no competing interests.

References

1. Ballestra LV, Guizzardi A, Palladini F (2019) Forecasting and trading on the VIX futures market: a neural network approach based on open to close returns and coincident indicators. *Int J Forecast* 35(4):1250–1262
2. Chaudhuri TD, Ghosh I (2016) Forecasting volatility in Indian stock market using artificial neural network with multiple inputs and outputs. [arXiv:1604.05008](https://arxiv.org/abs/1604.05008)
3. Choi J-E, Shin DW (2022) Parallel architecture of CNN-bidirectional LSTMs for implied volatility forecast. *J Forecast* 41(6):1087–1098
4. Chen W-J, Yao J-J, Shao Y-H (2023) Volatility forecasting using deep neural network with time-series feature embedding. *Econ Res-Ekonomska istraživanja* 36(1):1377–1401
5. Dixit G, Roy D, Uppal N (2013) Predicting India Volatility Index: an application of artificial neural network. *Int J Comput Appl* 70(4)
6. Jung G, Choi S-Y (2021) Forecasting foreign exchange volatility using deep learning autoencoder-LSTM techniques. *Complexity* 2021(1):6647534
7. Kumar P, Kumar S, Sharma R (2022) An impact of FPI inflows, nifty returns, and S&P returns on India VIX volatility. *World Rev Sci Technol Sustain Dev* 18(3–4):289–308
8. Kashif K, Ślepaczuk R (2025) LSTM-ARIMA as a hybrid approach in algorithmic investment strategies. *Knowl-Based Syst*, 113563
9. Liang Z, Ismail MT (2025) Advanced CEEMD hybrid model for VIX forecasting: optimized decision trees and ARIMA integration. *Evol Intel* 18(1):12
10. Liu Y (2019) Novel volatility forecasting using deep learning–long short term memory recurrent neural networks. *Expert Syst Appl* 132:99–109
11. Nti IK, Adekoya AF, Weyori BA (2020) A comprehensive evaluation of ensemble learning for stock-market prediction. *J Big Data* 7(1):20
12. Prasad A, Bakhshi P (2022) Forecasting the direction of daily changes in the India VIX index using machine learning. *J Risk Financial Manag* 15(12):552
13. Prasad A, Bakhshi P, Guha D (2023) Forecasting the direction of daily changes in the India VIX index using deep learning. *IIMB Manag Rev* 35(2):149–163
14. Pranav B, Hegde V (2021) Volatility forecasting techniques using neural networks: a review. *Int J Eng Res Technol (IJERT)* 10:748–52
15. Roszyk N, Ślepaczuk R (2024) The hybrid forecast of S&P 500 volatility ensembled from VIX, GARCH and LSTM models. [arXiv:2407.16780](https://arxiv.org/abs/2407.16780)

16. Singh P, Jha M (2024) Portfolio optimization using novel ew-mv method in conjunction with asset preselection. *Comput Econ*, 1–30
17. Singh P, Jha M, Sharaf M, Elmeligy MA, Gadekallu TR (2023) Harnessing a hybrid CNN-LSTM model for portfolio performance: a case study on stock selection and optimization. *IEEE Access*
18. Sharma A, Verma C (2024) Investigating the impact of technical indicators on option price prediction through deep learning models. In: 2024 IEEE International Conference on Electronics, Computing and Communication Technologies (CONECCT). IEEE, pp 1–5
19. Sharma A, Verma CK, Singh P (2024) Enhancing option pricing accuracy in the Indian market: a CNN-BiLSTM approach. *Comput Econ*, 1–28
20. Xiong R, Nichols EP, Shen Y (2015) Deep learning stock volatility with google domestic trends. [arXiv:1512.04916](https://arxiv.org/abs/1512.04916)
21. Zhao K, Zhang J, Liu Q (2022) Dual-hybrid modeling for option pricing of CSI 300ETF. *Information* 13(1):36

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